**Recruiting Team**
*Jane Street*
*250 Vesey Street*
*New York, NY 10281*

February 17, 2026

Dear Recruiting Team,

I am drawn to Jane Street's ML Engineering Internship because of something your description names directly: the gap between textbook ML and its application to noisy financial data. That gap is where I have spent most of my time, and it is where I find the most interesting problems.

At Columbia, I worked with Director of Financial Engineering Ali Hirsa on latent factor models for tabular financial data, where the core challenge was not just building models whose predictions generalize across market regimes, but ensuring their explanations remain faithful when conditions shift. I also worked with Director of Electrical Engineering Zoran Kostic on DeepSeek's GRPO algorithm.

Before Columbia, I spent five years at the Texas Advanced Computing Center orchestrating distributed ML and simulation workloads across leadership-class supercomputers, teaching graduate courses in ML, CUDA, and C++, and serving as a technical reviewer for $10M+ in NSF compute allocations.

I also co-founded Homecastr, a real-estate pricing platform where I shipped ML end-to-end. Property records are sparse, noisy, and non-stationary – much like financial data. Designing backtested walk-forward multi-horizon evaluation taught me more about rigorous ML engineering than any course.

My formal background is a B.S. in Pure Mathematics, and I have taught graduate courses in ML, CUDA, and C++ – but I learn fastest in environments where the feedback is real and fast, which is exactly what draws me to Jane Street. A full list of publications is on my Google Scholar; the attached resume has the details.

Sincerely,

**Daniel Hardesty Lewis**

*Attached: curriculum vitæ*

**Daniel Hardesty Lewis**
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# Daniel Hardesty Lewis

*ML Engineer – training, inference & evaluation on noisy real-world data*

ML engineer specializing in training, inference, and evaluation on noisy real-world data. Shipped financial factor models, real-estate pricing systems, and distributed HPC workflows at supercomputer scale.

## Core Expertise

8 years **ML Engineering**: Build and operate ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

## Technologies

| | |
|---|---|
| Modeling | PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT |
| LLM apps | embeddings; RAG; LangChain; LangGraph; DSPy; FAISS |
| Data stack | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase |
| Infrastructure | Linux/Unix; Bash; Git; Docker; Kubernetes; Ray; Spark; Dask; Airflow; CI/CD; observability |
| Additional | CUDA; C++; TypeScript; React |

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation; owned end-to-end workflow from source data to web delivery.

- Deployed an NYC semi-supervised VAE current-price model using **tax and sales records** to handle **sparsity and noise**; achieved 12% holdout error (vs. Zillow's 8.4% internal apples-to-apples).
- Shipped a Houston diffusion-based forecasting system with fan charts; achieved as low as ∼8% annualized compounding error, balancing design choices of error vs. trajectory realism, delivered via production web UI: homecastr.com
- Refined product positioning and execution milestones via strategic feedback from a16z, MetaProp, and others.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Directed development of a **conversational AI interface** simplifying global trade workflows; enabled decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with the founder of ScoutyAI and tariff economists at Columbia Business School to validate product-market fit and pricing assumptions.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Developed the web distribution platform end-to-end including data pipeline, tiling, hosting, delivery UX to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Co-founded summitgeospatial.com to commercialize and sustain state-level terrain models developed for the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, orchestrating 800,000-node distributed jobs across containerized HPC clusters while managing multi-million-dollar compute budgets.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

## Research

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% $R^2$ vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Math benchmarking and finetuning of Deepseek's GRPO algorithm

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

## Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

# Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

# Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

# Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

# Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

# Languages

Spanish Native Proficiency

French Professional Working Proficiency

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